The Family and School Connectedness Scales

Description and purpose

The *Family* and *School Connectedness Scales* assess two potential protective factors for children: connectedness with family and connectedness to school. They indicate the extent to which particular adolescents have a supportive network of responsible adults. The scales also measure self-reported happiness, belonging, safety and closeness to others at school, as well as fair treatment by teachers. Developed for the US National Longitudinal Study of Adolescent Health (Resnick et al. 1997) these measures have been used to research the relationship between schooling and health.

In an analysis of data on over 12,000 young people aged 12 to 18 years, Resnick et al. (1997) found important protective effects of perceived family and school connectedness on a wide range of health-risk behaviours. These included self-reported emotional distress; suicidal thoughts and behaviour; violence; use of cigarettes, alcohol and marijuana; and age of first sexual experience. The more strongly the young people felt a sense of connection to family and school, the less likely they were to engage in each of these health-risk behaviours. In addition to low school connectedness, low attainment at school was associated with high risk, leading to strong support for the concept of the 'full service school' as a community-based point of delivery of education, social care and health services to address the range of needs experienced by vulnerable young people.

The *Family* and *School Connectedness Scales* are therefore highly relevant to the evaluation of extended schools initiatives, parental engagement/involvement projects, and any other intervention designed to increase the connection of children and young people to family or school. For example, the scales have been used successfully in a controlled evaluation of a multidisciplinary mentoring programme for pupils aged 9 years. Significant improvements were found in school and family connectedness as well as self-esteem levels, depression and involvement in bullying and fighting (King et al. 2002).

Particular concerns are often expressed that young people from ethnic-minority backgrounds are likely to experience more barriers to forming a strong and positive connection to school due to the cultural and linguistic differences between their school, family and community contexts. Dornbusch et al. (2001) found that adolescent attachments to family and school tended to reduce the overall frequency, prevalence and intensity of involvement in a range of deviant behaviour, regardless of community context, gender, or ethnic group. Their findings indicated that ties to the school appeared to operate preventatively so as to deter adolescents from engaging at all in a specific form of deviation. Low attachment to school may help create opportunities to engage in problematic behaviour, for example through truancy or association with peers who are similarly disconnected from school or already involved in deviant behaviour. Similar findings were reported in Henrich, Brookmeyer and Sharhar's (2005) study of weapon violence in adolescence, where family and school connectedness seemed to have more power as preventative rather than as mitigating factors. The authors consider that parents who maintain close relationships with their children may be more effective at monitoring and controlling how, with whom and where adolescents spend their time, so reducing the chance of exposure to violence. However, once the cycle of violence has escalated to the point where their children use weapons, parents may have much reduced opportunity for intervention.

Administration

The *Family Connectedness Scale* consists of 12 items adapted from the US National Longitudinal Study of Adolescent Health (Resnick et al. 1997). The items assess the quality of the relationships adolescents report having with their parents. The first eight questions are rated on a five-point scale ranging from *'Not at all'* (1) to *'Very much'* (5). The next four items are rated on a five-point scale ranging from *'Strongly agree'* (5) to *'Strongly disagree'* (1).

The *School Connectedness Scale* consists of six items adapted from the US National Longitudinal Study of Adolescent Health (Resnick et al. 1997). The items are rated on a five-point scale ranging from 5 'Strongly agree' to 1 'Strongly disagree'.

The scales can be administered to whole classes, groups or individuals. It is important to inform pupils at the start of the session about the purpose of the assessment and the uses that may be made of the information. Assurances about confidentiality, where appropriate, should be given. A young person may respond very differently to the *Family Connectedness Scale* if they think it might be shown to their parents than if they are assured that the results will be confidential to professionals working with them in school.

The scales may be administered as part of a longer whole school or year group survey. This is the way in which they were administered in the US national longitudinal study for which they were developed.

Scoring

The scores on each item are summed for each scale. This gives a total score on the 12-item *Family Connectedness Scale* of between 12 and 60 and a total score on the six-item *School Connectedness Scale* of between 6 and 30.

Interpretation

The Connectedness Scales scores have most often been used to look broadly at associations between school and family connectedness and known risk factors, including the identification of changes across groups or schools following intervention. In evaluating the effects of school development or parental involvement initiatives it will be important to administer the relevant questionnaire prior to the start of the intervention to provide a baseline and after the intervention has been running long enough that an effect could reasonably be expected. In comparing changes in the scores over time, the application of statistical tests is recommended to evaluate the significance of the results obtained. Schools may want to consult their educational psychology service for assistance with this.

At present there are no normative or national data in the UK for these scales. However, in one of the few studies to interpret the scores individually, Ozer et al. (2008) provide guidance about *School Connectedness* scores that could be categorised as 'Low', 'Midpoint' and 'High'. These are shown in Table 1.6, with a parallel set of categories for the *Family Connectedness Scale* based on the same principles. However, it should be noted that mean scores on the *Family Connectedness Scale* tend to be higher on average than scores on the *School Connectedness Scale* (e.g. *Family Connectedness Scale*: M = 4.11 sd = .60, *School Connectedness Scale* M = 3.77 sd = .76, from Meuller 2009). These score categories are offered tentatively, not as diagnostic indicators but as a starting point for further exploration with young people targeted as a result of concern raised by low scores. The following case study, adapted from Ozer et al. 2008, is provided as an example:

> *Grace's parents were 'poor' in Hong Kong, and moved to the United States to provide opportunities for their children. While in middle school Grace had scored in the mid-point range on the* School Connectedness Scale*; in high school she scored low. Grace expressed negative feelings about high school, citing pushing and sexual harassment in the hallways. She reported that her middle school was safer than her high school, where she felt she was targeted in the hallways because non-Chinese students assumed that she was easy to push around. Although in the past Grace was an excellent student, her grades declined towards the end of high school after her father was diagnosed with cancer. Grace reported feeling unhappy at school and experiencing a great deal of stress due to her father's illness. She did, however,*

Table 1.6 Guidance on score categorisation

	Scores categorised as LOW	Scores categorised as MID-POINT	Scores categorised as HIGH
Family Connectedness Scale	12–36	37–47	48–60
School Connectedness Scale	6–18	19–23	24–30

Source: Ozer et al. 2008

report several meaningful relationships with teachers who provided her with emotional and practical support. She reported a particularly close relationship with her gym teacher who knew about her father's illness. Like most students who discussed a personal problem with a teacher, however, Grace did not initiate the discussion. When asked how her teacher learned about her father's illness, she replied, 'I think he asked me why I was absent so much'. Grace's narrative illustrates a theme identified earlier – how a teachers' inquiry into why she was absent provided an opportunity for a deepening of the relationship – and it also highlights how student–student relationships (e.g. perceived harassment) can undermine sense of connectedness to school.

Technical details

Sieving et al. (2001) present detailed information on the construction of the Connectedness Scales. They report internal consistency values across gender and racial/ethnic groups for a large representative sample of twelve- to eighteen-year-olds in the US of .81 to .85 for family connectedness and .75 to .82 for school connectedness.

In support of the construct validity of the scales, Sieving et al. (2001) report significant correlations of .35 to .45 with measures of psychological wellbeing, for example emotional distress and self-esteem. There is also evidence that positive classroom management climates, participation in extra-curricular activities and tolerant discipline policies are associated with higher levels of school connectedness (McNeely et al. 2002). Loukas et al. (2006) found that pupil-perceived school climate (cohesion, friction, competition among students and overall satisfaction with classes as assessed by the My Class Inventory[4]) facilitated connectedness to school and that pupils who reported being more connected to school subsequently engaged in fewer conduct problems (as assessed by the *Strengths and Difficulties Questionnaire*; Goodman 1997) than did their less connected peers.

Gay, lesbian, bisexual, transgendered and questioning (GLBTQ) youth are a group whose school connectedness may be at particular risk as there is evidence that school environments are all too often a source of peer harassment, bullying and aggression. Galliher et al. (2004) found that adolescents who reported same sex or both sex attraction scored lower on the *School Connectedness Scale* than the majority (94.7 per cent) who reported opposite sex attraction. Other minority groups have also been found to report significantly lower-school connectedness, for example Icelandic ten- to twelve-year-old children with a mental illness or learning disabilities were found to report significantly lower-school connectedness than their school mates who had physical chronic illness(es) and they in turn scored lower than those who were typically developing (Svavarsdottir 2008). It is suggested that schools should be alert to the need to take preventive action in order to reduce the risk of disengagement for vulnerable groups.

[4]The 'My Class Inventory' can be found in 'The Learning Environment' booklet of *Psychology in Education Portfolio* (GL Assessment 1999).

Comparison with similar measures

Because they were developed for the US National Longitudinal Study of Adolescent Health, the *Family* and *School Connectedness Scales* are the measures of connectedness on which there is the most extensive data. In contrast to the single-factor *School Connectedness Scale*, Brown and Evans (2002) have developed a multi-component measure comprising four hypothesised elements of connection: commitment, power, belief and belonging. Each element is assessed by four items, giving 16 items in total.

A measure of family connectedness developed by Hardway and Fuligni (2006) consists of multiple measures of family relationships and daily behaviour, designed to provide a rich portrait of family connectedness among adolescents across different ethnic backgrounds. A questionnaire collects information on respect for family, perceived obligations to provide current and future assistance, family identity and closeness to father and mother. In addition a daily checklist measure assesses helping the family, spending leisure time with the family and time spent studying and socialising with friends.

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